Digital Egypt Pioneers

Sustainability

وزارة الاتصـــــالات
وتكنولوجيا المعلومات

SUSTAINABILITY ANALYSIS REPORT

UNDER SUPERVISION OF
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SUBMITTED TO: DEPI

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NOOR MAGED

Contents :

1. Abstract:

- This project presents a comprehensive data analytics pipeline that integrates SQL, Python, Power BI, and Tableau to extract, process, analyze, and communicate actionable insights from a real-world dataset. First, raw data was ingested and cleaned using Python employing libraries such as Pandas for data transformation, handling missing values, and performing necessary preprocessing. Next, structured data was stored in a relational database and queried using SQL to perform in-depth analysis, aggregation, and to build meaningful relationships among data entities.
- Following the data preparation and analysis stages, two interactive dashboards were developed using Power BI and Tableau. These dashboards enable stakeholders to explore key performance indicators, trends, and patterns dynamically. Power BI was used to create business-focused dashboards with drill-down capabilities, while Tableau provided visually rich, story-driven visualizations that highlight temporal trends and comparisons.
- Through this multi-tool approach, the project demonstrates how combining programmatic data cleaning (Python), relational data modeling and querying (SQL), and business intelligence visualization tools (Power BI and Tableau) can drive data-driven decision-making. The results include clear insights into the dataset's major drivers, opportunities for optimization, and recommendations for future strategic actions.

2. Introduction:

2.3 Project Overview:

Global Sustainability Performance

- This repository serves as the culminating portfolio for the **Digital Egypt Pioneers (DEPI) Initiative**, undertaken under the direct supervision of the **Ministry of Communication and Information Technology (MCIT)**.
- This project delivers a complete, end-to-end analysis of a global Product Sustainability Performance Dataset. By leveraging a unified dataset across five distinct, industry-standard tools, this portfolio demonstrates mastery over the entire modern data analytics stack, from robust data engineering to advanced visual storytelling.
- In an era where environmental consciousness is no longer optional but a strategic imperative, this project evaluates the sustainability performance of consumer products across global markets. The objective of this analysis is to quantify the environmental impact of manufacturing across distinct metrics: **Carbon Footprint**, **Water Usage**, and **Waste Production**.

2.4 Scope:

The analysis utilizes a comprehensive dataset linking manufacturing facts with dimensions such as Brand, Material Type, Region, and Market Trends. The technical execution of this project leveraged a robust, multi-tiered technology stack:

- **Data Processing & Modeling: SQL** was employed for rigorous data extraction and relational structuring, while **Excel** served as the primary tool for preliminary data validation and cleaning. **Python** was utilized for advanced statistical analysis and correlation modeling to identify deep-seated trends in sustainability performance.
- **Visualization & Reporting:** Strategic insights were crystallized using **PowerBI** and **Tableau**. These tools provided interactive geospatial and performance dashboards, enabling a granular view of regional disparities and material efficiency.

The data model integrates:

- **Performance Metrics:** Quantitative analysis of carbon emissions (kg CO_2e), water consumption (liters), and waste generation.
- **Categorization:** Evaluation of products across "Fully Sustainable," "Partially Sustainable," and "Not Sustainable" classifications.
- **Geographical Focus:** A comparative study of production impacts across Africa, the Americas, Asia, and Europe.
- **Material Insights:** Assessing the renewability of inputs ranging from Organic Cotton and Bamboo to Recycled Plastics and Stainless Steel..

3. The Analytical Goal:

- The core objective was to transform raw, complex data on environmental metrics (Carbon Footprint, Water Usage, Waste Production) and business factors (Price, Brand, Material Type) into **actionable insights** that quantify the relationship between corporate practice and sustainability outcomes.

4. Multi-Tool Workflow: The 5-Step Data Journey:

4..1.Data Engineering & Modeling (SQL):

- **Focus:** Data integrity, standardization, and performance optimization.
- **Techniques:** Used advanced SQL to perform rigorous data cleaning, text standardization, and missing value management. The raw data was meticulously transformed into an analytical **Star Schema** (Fact and Dimension Tables) to ensure high-performance querying for subsequent tools. **Outliers were managed** using the Interquartile Range (IQR) method (Winsorizing) to ensure statistical accuracy.
- **Deliverable:** A cleaned, normalized, and performant SQL database (Sustainability_db) ready for BI consumption.

4..2.Statistical Analysis & Transformation (Python):

- **Focus:** Deep statistical exploration and programmatic data manipulation.
- **Techniques:** Employed the **Pandas** library for robust data cleaning, restructuring, and feature engineering. **NumPy** was used for statistical validation, while **Matplotlib** and

Seaborn were utilized to generate initial visualizations and uncover underlying distributions and correlations.

- **Deliverable:** Documented statistical findings and validated data preparation steps, ready for visualization.

4..3.Rapid Reporting & Metric Calculation (Microsoft Excel):

- **Focus:** Agility, fast aggregation, and executive reporting preparation.
- **Techniques:** Utilized **Pivot Tables** and **Pivot Charts** for multi-dimensional analysis. Complex lookups and conditional formatting were applied to extract and highlight key performance indicators (KPIs) quickly, demonstrating proficiency in corporate reporting standards.
- **Deliverable:** Dynamic Pivot-based reports providing immediate answers to critical business questions.

4..4.Enterprise Business Intelligence (Power BI):

- **Focus:** Creating professional, scalable, and interactive dashboards.
- **Techniques:** Built a robust Data Model, defined optimal table relationships, and authored complex custom metrics using **DAX (Data Analysis Expressions)**. Developed interactive dashboards with filters and drill-through capabilities, ensuring a user-friendly BI experience.
- **Deliverable:** An interactive, analytical Power BI dashboard allowing users to explore sustainability performance by region, brand, and material type.

4..5.Advanced Visual Storytelling (Tableau):

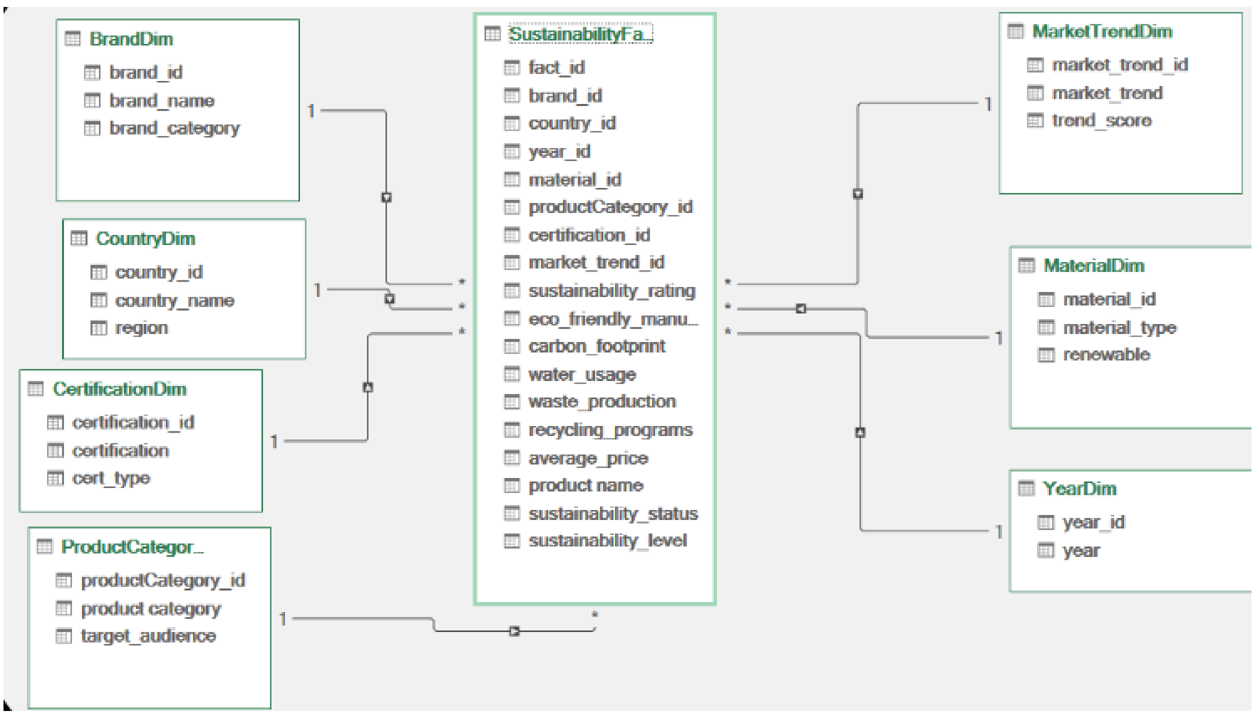
- **Focus:** Aesthetically superior visualization and advanced visual analytics.
- **Techniques:** Created high-impact visual reports, leveraging Tableau's unique capabilities for geographic mapping, scatter plots with trend lines, and parameter-driven analysis. Focused on data storytelling to effectively communicate core findings.
- **Deliverable:** A visually compelling Tableau dashboard that highlights key market trends and top/bottom performers in sustainability.

5. Key Questions Answered:

- How do environmental certifications and recycling programs impact a
- product's overall sustainability rating?
- Which brands and regions lead in eco-friendly manufacturing?
- What is the correlation between product price and sustainability score?
- How can data models be optimized for rapid, complex reporting?

6. Methodology:

6.1 Modelling:



i. Data Model Description: Sustainability Analysis Schema

The image displays a **dimensional data model**, which is typically used for Business Intelligence (BI) and reporting. It's structured as a **Star Schema**, where a central **Fact Table** is connected to multiple surrounding **Dimension Tables**. This model is designed to efficiently analyze all aspects of the sustainability project.

Central Fact Table

- **SustainabilityFact:** This is the core table where the **measurable data** (facts) and **foreign keys** connecting to the dimensions are stored.
  - **Measures (Metrics):** This table contains the key sustainability performance indicators (KPIs) that are being analyzed:
    - sustainability_rating

- eco_friendly_manu... (Manufacturing?)
- carbon_footprint
- water_usage
- waste_production
- recycling_programs
- average_price
- sustainability_status
- sustainability_level

- **Keys:** It holds foreign keys linking it to all dimension tables (e.g., brand_id, country_id, year_id, material_id, etc.).

Dimension Tables (Context)

The Dimension tables provide the **context** (who, what, when, where, why) for the measures in the Fact Table. They are all connected to SustainabilityFact in a **One-to-Many (1-*)** relationship.

Dimension Table	**Attributes (Context Provided)**	**Purpose in Sustainability Project**
BrandDim	brand_id, brand_name, brand_category	Identifies the specific **company/brand** associated with the sustainability data.
CountryDim	country_id, country_name, region	Provides the **geographical location** where the sustainability data was recorded.
CertificationDim	certification_id, cert_type	Categorizes the specific **sustainability certifications** (e.g., Fair Trade, ISO14001, etc.) a product or company holds.
ProductCategoryDim	productCategory_id, product_category, target_audience	Describes the **type of product** for which the metrics apply.
MarketTrendDim	market_trend_id, market_trend, trend_score	Allows analysis of how sustainability metrics correlate with **external market factors or trends**.

Dimension Table	**Attributes (Context Provided)**	**Purpose in Sustainability Project**
MaterialDim	material_id, material_type, renewable	Provides details on the **raw materials** used (e.g., organic cotton, recycled plastic) and whether they are renewable.
YearDim	year_id, year	Provides the **time dimension**, allowing for **longitudinal analysis** and tracking changes over time.

Significance for the Project Document

This data model confirms that your project is built on a **robust analytical foundation** capable of answering complex questions, such as:

- "How has the **carbon footprint** (carbon_footprint) changed over the last **five years** (YearDim) for **brands** (BrandDim) operating in the **European region** (CountryDim)?"
- "What is the average **waste production** (waste_production) for products using **renewable** (MaterialDim) materials compared to non-renewable ones?"
- "What **certification** (CertificationDim) is most strongly correlated with a high **sustainability rating** (sustainability_rating)?"

This model is a blueprint for your project's data-driven conclusions about sustainability.

6.2 Charts:

i. Chart Description: Certifications and Waste Reduction

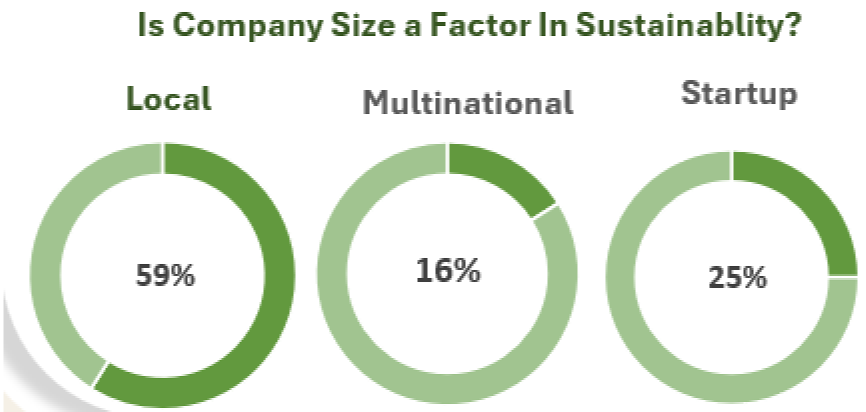
This double bar chart investigates the question: **"Do Certifications Really Make a Difference in Reducing Waste?"** by comparing two metrics across eight sustainability certifications and a "None" category.

Metric	Color	Description
Average of sustainability_rating	Dark Green	An assessed score of sustainability commitment.
Average of waste_production	Light Green	The measured amount of waste produced.

Key Findings for Sustainability Documentation

1. **General Trend:** Across all categories, the **Average of waste_production** (Light Green) is significantly **higher** than the **Average of sustainability_rating** (Dark Green). This suggests that a high sustainability rating alone does not directly correlate with a low waste output.
2. **Impact of Certification:** The **"None"** category (no certification) exhibits the **highest** level of **waste production** (approaching 8.00). This is the strongest evidence provided by the chart that **sustainability certifications, as a whole, are associated with reduced waste** compared to having no formal certification.
3. **Top/Bottom Performers:**
    - **Eu Ecolabel** shows the lowest level of waste production (just over 6.00).
    - **Iso14001** and **Gots** show the highest waste production among the certified groups.

4. **Overall Conclusion:** The data supports the conclusion that while certified companies still face waste challenges, **formal certification is a key differentiator** for lowering waste metrics relative to uncertified operations.



ii. Chart Description: Company Size and Sustainability

This visualization, titled **"Is Company Size a Factor In Sustainability?"**, uses three doughnut charts to show how different types of companies contribute to a measured sustainability metric (likely the sample size or overall influence in the dataset).

Chart Structure

The chart compares the contribution of three company size/type categories: **Local, Multinational, and Startup**.

- The **percentage value** represents the proportion of each company type in the data sample.
- The **darker green slice** appears to represent the actual percentage value (the majority of the circle for "Local," and smaller slices for the others).
- The **lighter green slice** represents the remaining percentage of the full circle.

Key Findings

The chart reveals a significant imbalance in the contribution of different company types to the dataset or the sustainability factor being measured:

1. **Local Companies Dominate: Local** companies account for the largest proportion at **59%**. This suggests that local businesses are the most numerous or are the most frequent contributors/focus in the study.
2. **Startups are Significant: Startups** make up the second-largest portion at **25%**.
3. **Multinationals are the Smallest Group: Multinational** corporations, despite their potential global impact, represent the smallest proportion at only **16%**.

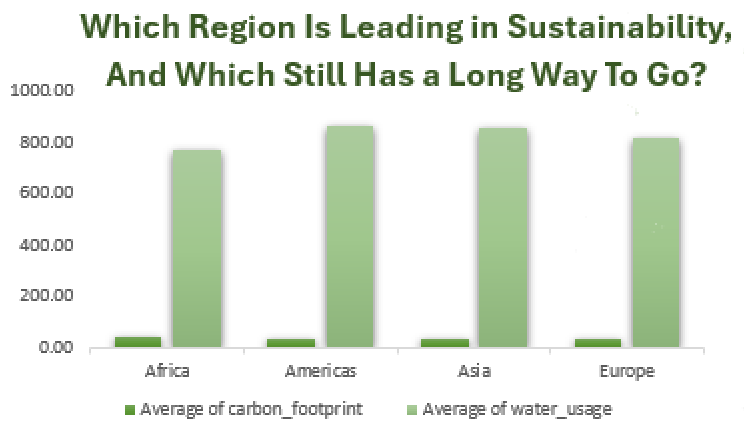
Interpretation for Sustainability Documentation

- This chart is crucial for defining the **scope and context** of your project's findings. If your overall sustainability conclusions are based on this sample, they will be heavily influenced by the performance of **Local** companies (59%).
- Any conclusions drawn about **sustainability practices** in your document will reflect the actions and data of **Local** businesses much more than Multinational ones.

- The small representation of Multinationals (16%) means that the study's conclusions about these large organizations might not be as statistically robust or representative of the whole sector.

iii. Chart Description:

- **Type:** Line Chart.
- **Title: "Is a Growing Market Also a Greener One?"**
- **Axes:** The X-axis classifies the **Market State** (**Declining, Rising, Stable**). The Y-axis tracks a **Footprint Metric** (ranging from 1610 to 1710).
- **Key Findings:** The **Rising** market state is clearly associated with the **highest** footprint metric (approximately 1695). The **Stable** market state corresponds to the **lowest** footprint metric (approximately 1640).
- **Interpretation:** The data suggests an inverse relationship between market growth and environmental performance; that is, a rapidly growing market is associated with a larger (less "green") environmental footprint compared to a stable or declining market.



iv. Chart Description: Regional Sustainability Comparison

Which Region Is Leading in Sustainability, And Which Still Has a Long Way To Go?

- **Type:** Double Bar Chart comparing two metrics by region: Average of carbon_footprint (Dark Green) and Average of water_usage (Light Green).
- **Regions Compared:** Africa, Americas, Asia, and Europe.
- **Key Finding 1 (Carbon Footprint):** All four regions show a **very similar and very low average carbon footprint** (all below 50). This suggests either successful reduction across the board or a low level of contribution from these regions in the sample.
- **Key Finding 2 (Water Usage Dominance): Water usage** metrics (Light Green bars) are the dominant environmental factor shown, with values around 800-900.
- **Key Finding 3 (Highest Water Usage):** The **Americas** and **Asia** show the highest average water usage (approaching 900), indicating these regions face the greatest challenges or have the highest resource intensity related to water.
- **Key Finding 4 (Lowest Water Usage): Africa** and **Europe** show slightly lower, but still high, average water usage (around 780-820), suggesting a marginal lead in water conservation efforts compared to the other regions.
- **Conclusion:** The sustainability challenge is focused on **water consumption** rather than carbon footprint, with the Americas and Asia showing the greatest water impact in this dataset.

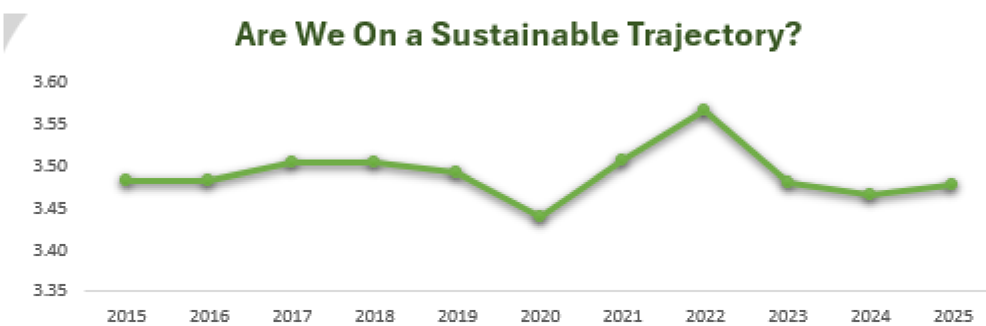
v. Chart Description: Global Recycling Footprint

Is Our Global Recycling Footprint Expanding Evenly?

- **Type:** Single Bar Chart (Regional Comparison).
- **X-Axis (Regions):** Africa, Americas, Asia, and Europe.
- **Y-Axis (Metric Value):** Represents a measure of recycling footprint, with values ranging up to 2500. The specific unit or metric is not labeled, but it clearly indicates the magnitude of recycling activity or capacity.
- **Key Finding 1 (Uneven Expansion):** The expansion of the global recycling footprint is **highly uneven**. Europe shows the largest footprint (reaching 2000), while Africa shows the smallest (less than 500).
- **Key Finding 2 (Leading Regions): Europe** has the largest recycling footprint, followed by **Asia** (around 1700).
- **Key Finding 3 (Lagging Regions):** The **Americas** (around 900) and especially **Africa** (below 500) show significantly smaller recycling footprints, suggesting a considerable need for investment or development in these regions to match the scale of Europe and Asia.
- **Conclusion:** This chart strongly demonstrates **regional disparity** in recycling capabilities, highlighting the areas where the "recycling footprint" needs to be expanded most significantly for equitable global sustainability progress.

vi. Choropleth Map Description: Environmental Leadership

- **Type:** Choropleth Map (A world map using color-coding to display geographical data).
- **Title: "Which Country Is The True Environmental Leader?"**
- **Metric:** The map displays a single metric labeled "Series 1," which is color-coded on a gradient from 373 (light green) to 455 (dark green). A higher number represents the value being measured (likely a form of environmental rating or footprint).
- **Key Findings:**
  - **Brazil** shows the highest value at **455**, indicated by the darkest color.
  - **India** (425) and **China** (424) also display significantly high values.
  - The **United States** has a value of **404**.
- **Interpretation:** Based solely on the magnitude of the "Series 1" metric, Brazil is quantitatively identified as the country with the highest recorded value.



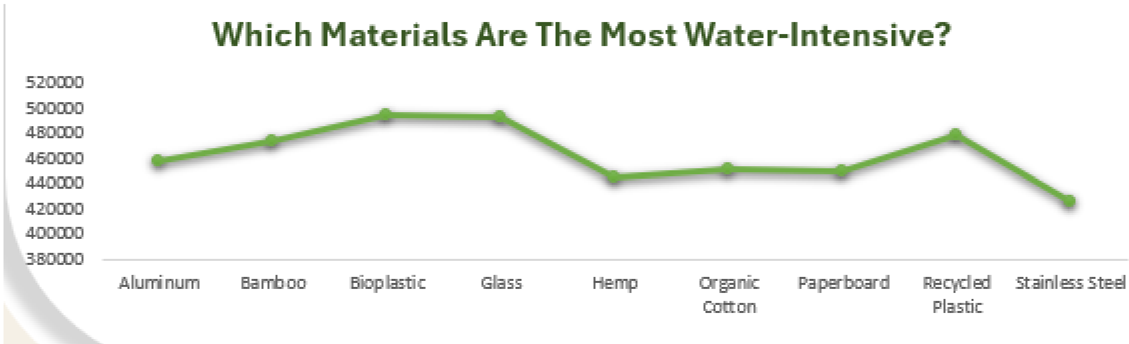
vii. Line Chart Description: Sustainable Trajectory (2015-2025)

- **Type:** Line Chart (Time Series).
- **Title: "Are We On a Sustainable Trajectory?"**
- **Axes:**
  - **X-axis (Time):** Shows years from **2015 to 2025**.
  - **Y-axis (Metric):** Displays a quantitative value (likely a sustainability score or index) ranging from 3.35 to 3.60.

- **Key Findings:**
    - The value remained relatively stable from 2015 to 2019 (around 3.47 to 3.50).
    - There was a noticeable **dip in 2020** (reaching the lowest point around 3.44).
    - The value rebounded strongly, peaking in **2022** (reaching the highest point at approximately 3.57).
    - The trajectory has been **declining since 2022**, ending in 2025 at a value of approximately 3.48.
- **Interpretation:** While the value has shown volatility (especially the 2022 peak and 2020 dip), the trajectory from 2022 to 2025 is downward, suggesting a current trend away from the previous high point, ending the 10-year period (2015-2025) at a level similar to 2019.

viii. Bar Chart Description: Product Category Value

- **Type:** Bar Chart.
- **Title: "Which Product Category Offers The Best Value?"**
- **Axes:**
    - **X-axis (Categories):** Lists 10 distinct product categories (e.g., **Clothing, Electronics Accessories, Garden, Office Supplies, Personal Care**).
    - **Y-axis (Metric):** Measures a value (likely a "Value" score or index) ranging from 28.00 to 32.00.
- **Key Findings:**
    - The **highest value** is offered by the **Office Supplies** category, reaching approximately 31.45.
    - The **second highest value** is the **Personal Care** category, at approximately 31.30.
    - The **lowest value** is in the **Home** category, at approximately 29.30.
    - **Clothing** also shows a relatively low value (approx. 29.75).
- **Interpretation:** The chart clearly identifies Office Supplies and Personal Care as the leading categories based on this specific "Value" metric, suggesting that projects focusing on these product areas may yield the highest measured return or benefit.



ix. Line Chart Description: Water-Intensive Materials

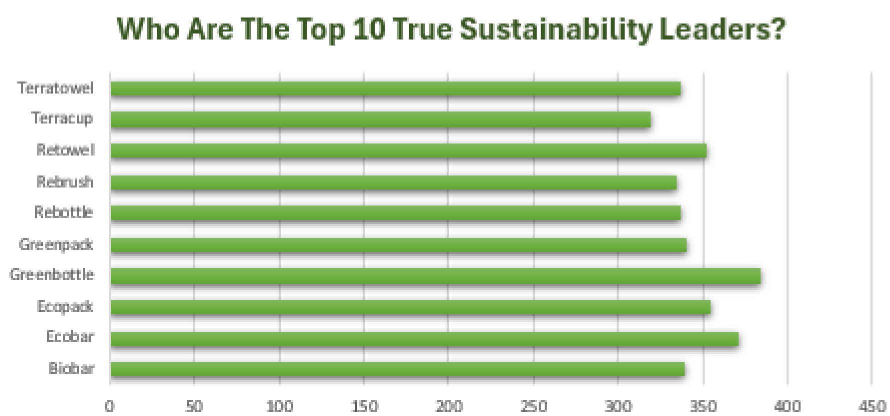
- **Type:** Line Chart.
- **Title: "Which Materials Are The Most Water-Intensive?"**
- **Axes:**
  - **X-axis (Material):** Lists nine materials (e.g., Aluminum, Bamboo, Bioplastic, Glass, Stainless Steel).
  - **Y-axis (Metric):** Measures the water intensity, with values ranging from 380,000 to 520,000.
- **Key Findings:**
  - **Highest Water Intensity: Bioplastic** and **Glass** share the highest value, both peaking around **495,000**.
  - **Lowest Water Intensity: Stainless Steel** shows the lowest intensity, at approximately **420,000**.
  - **Recycled Plastic** is also high, around **475,000**.
- **Interpretation:** This data is crucial for material selection, as it identifies Bioplastic and Glass as the most water-intensive options among those compared.

x. Chart Description: Sustainability Cost Analysis

- **Type:** Three Doughnut Charts, side-by-side.
- **Title: "Does Sustainability Cost a Lot?"**
- **Categories:** The chart presents three classifications: **Fully Sustainable (39%), Partially Sustainable (34%),** and **Not Sustainable (25%)**.
- **Key Findings:**
  - The largest single group is **Fully Sustainable** at **39%**.

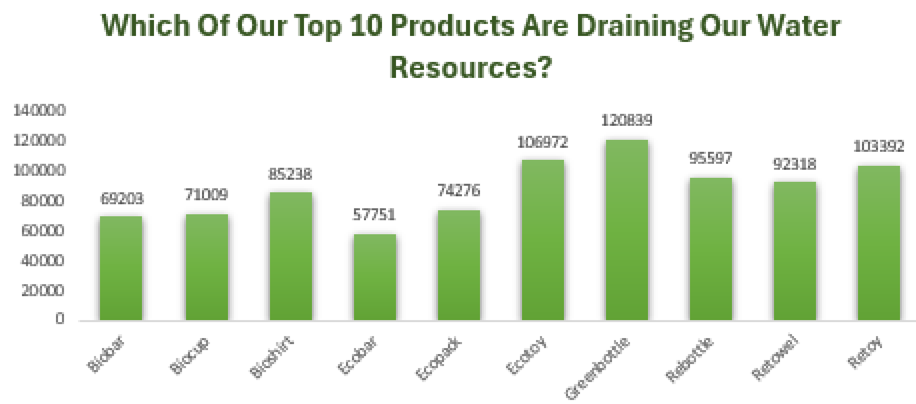
- Combined, the **Fully Sustainable** (39%) and **Partially Sustainable** (34%) categories account for **73%** of the total, suggesting that sustainability features are present in the vast majority of cases.
- The **Not Sustainable** category is the smallest at **25%**.

- **Interpretation:** The data indicates that the majority of items or instances measured are classified as either fully or partially sustainable, which suggests that sustainability is not prohibitive across the dataset, addressing the question posed by the title.



xi. Bar Chart Description: Top 10 Sustainability Leaders

- **Type:** Horizontal Bar Chart.
- **Title: "Who Are The Top 10 True Sustainability Leaders?"**
- **Axes:**
  - **Y-axis (Categories):** Lists 10 product names/brands (e.g., **Terratowel, Greenbottle, Ecobar, Biobar**).
  - **X-axis (Metric):** Measures a quantitative value (likely a "Sustainability Leader" score or index) ranging from 0 to 450.
- **Key Findings:**
  - **Highest Score: Greenbottle** achieved the highest score, near 380.
  - **Second Highest Score: Ecobar** achieved the second-highest score, near 370.
  - **Lowest Score (in Top 10):** Products like **Terratowel** and **Biobar** achieved the lowest scores in the top 10 list, both around 340.
  - **Tight Grouping:** The top 10 scores are very tightly grouped, all falling between approximately 340 and 380.
- **Interpretation:** The chart identifies the top 10 entities based on the measured sustainability metric, with Greenbottle and Ecobar clearly leading the group. The narrow range of scores indicates intense competition or similar performance levels among these leaders.

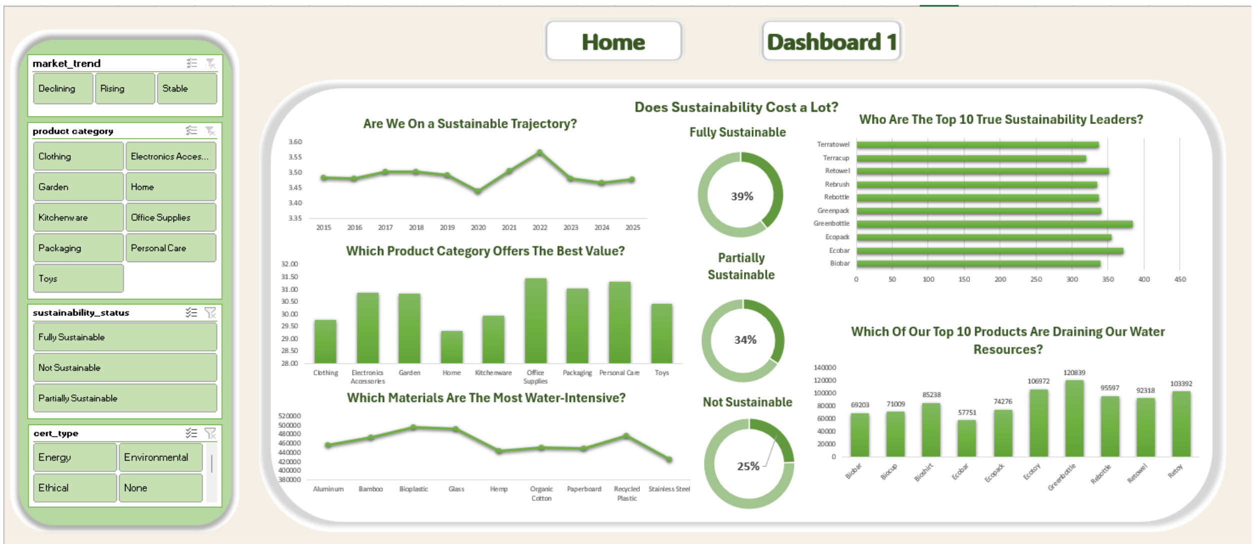


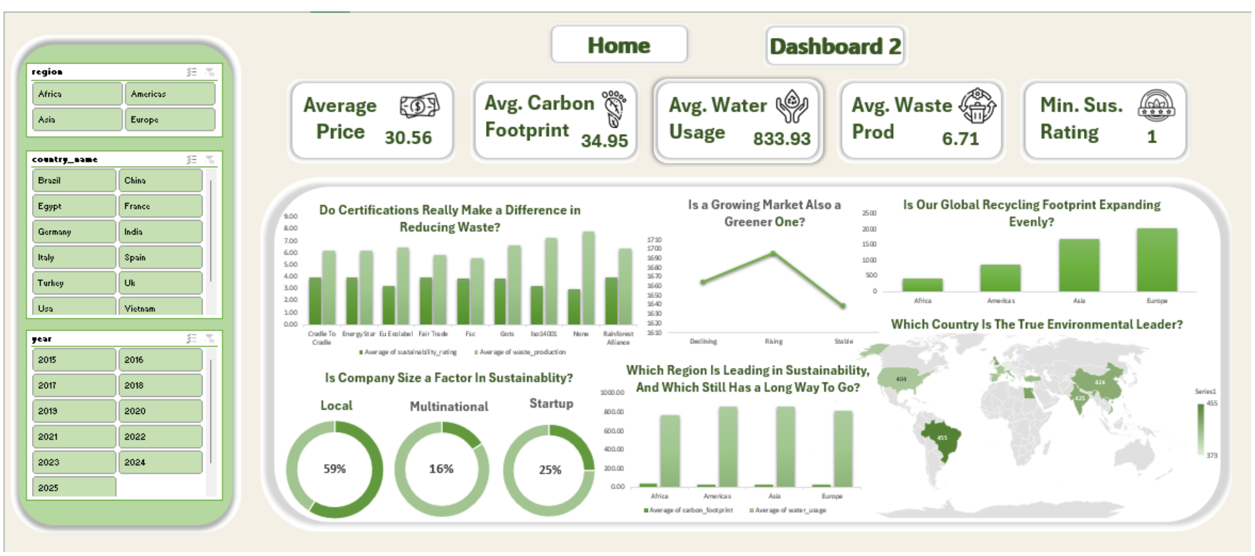
xii. Bar Chart Description: Top 10 Products Draining Water Resources

- **Type:** Vertical Bar Chart.
- **Title: "Which Of Our Top 10 Products Are Draining Our Water Resources?"**
- **Axes:**
  - **X-axis (Products):** Lists 10 specific product names (e.g., **Biobar, Biocup, Bioshirt, Ecobar, Greenbottle, Retoy**).
  - **Y-axis (Metric):** Measures a water usage metric (units ranging from 0 to 140,000).
- **Key Findings:**
  - **Highest Water Usage: Greenbottle** shows the maximum water usage at **120,839**.
  - **Second Highest Water Usage: Ecotoy** is the next highest at **106,972**.
  - **Lowest Water Usage: Ecobar** has the lowest usage among the top 10, at **57,751**.
  - **Biobar** and **Biocup** also show low usage (69,203 and 71,009, respectively).
- **Interpretation:** The data clearly identifies **Greenbottle** and **Ecotoy** as the products contributing the most to water resource depletion within the analyzed top 10, marking them as priorities for optimization efforts.

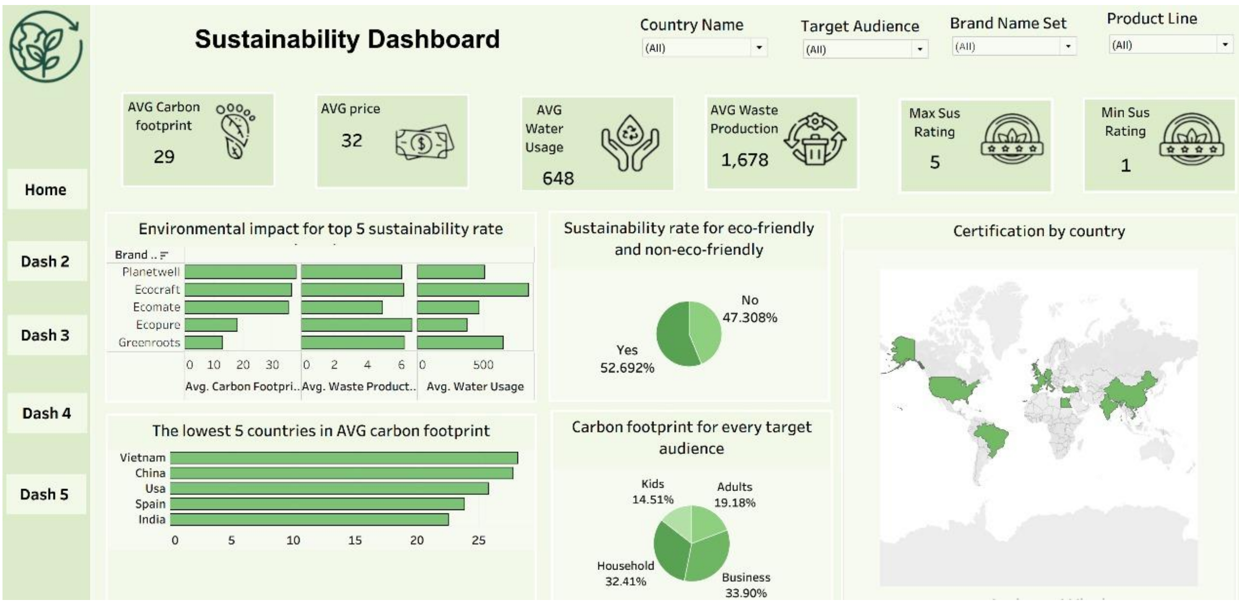
6.3 Dashboards:

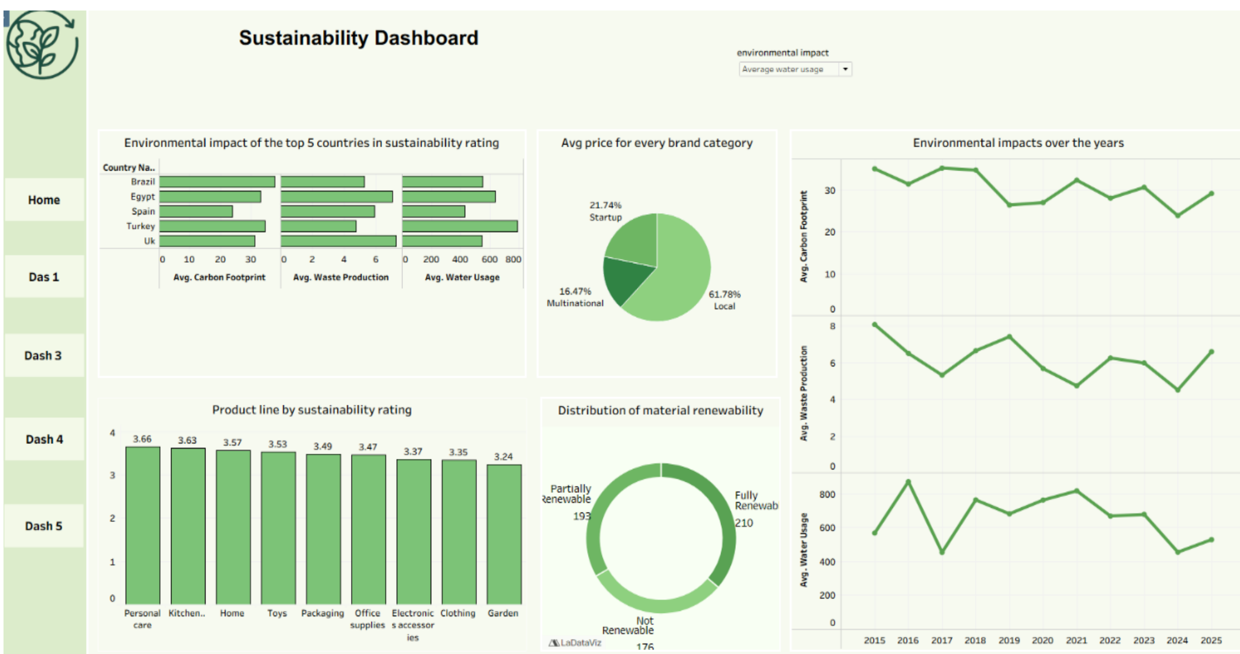
6.3.1 Excel:

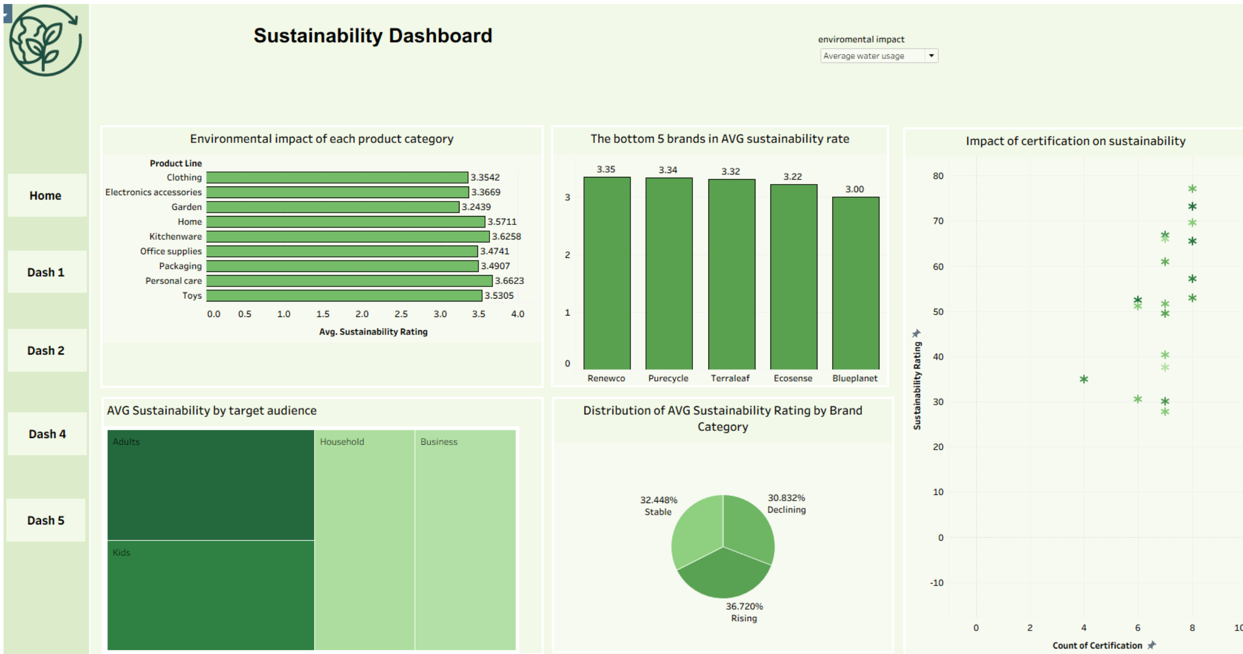


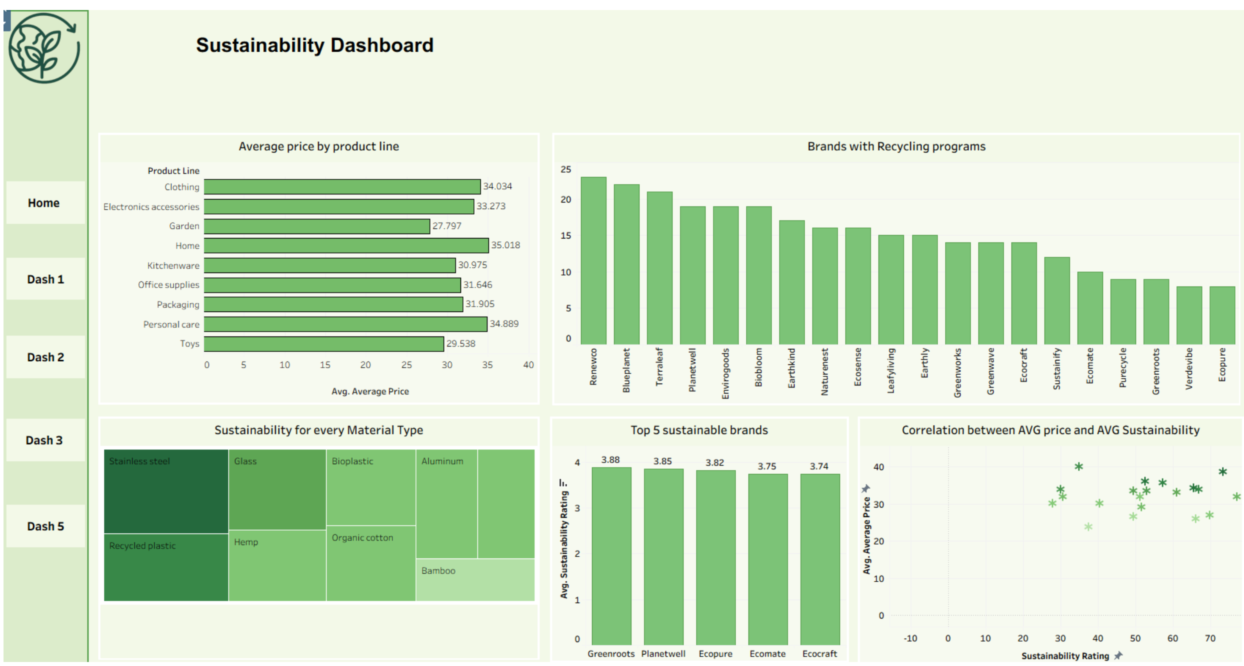


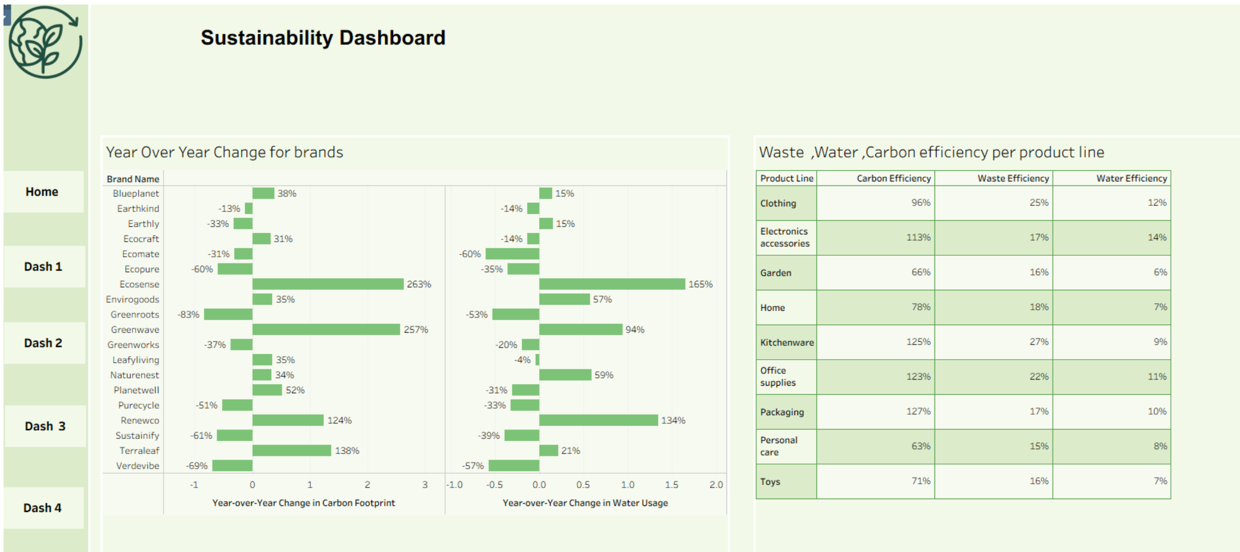
6.3.2 Tableau:


Sustainability Dashboard
Country Name
(All)
Target Audience
(All)
Brand Name Set
(All)
Product Line
(All)
AVG Carbon footprint
29
AVG price
32
AVG Water Usage
648
AVG Waste Production
1,678
Max Sus Rating
5
Min Sus Rating
1
Home
Dash 2
Dash 3
Dash 4
Dash 5
Environmental impact for top 5 sustainability rate
Brand ..
Planetwell
Ecocraft
Ecomate
Ecopure
Greenroots
Avg. Carbon Footpri..
Avg. Waste Product..
Avg. Water Usage
Sustainability rate for eco-friendly and non-eco-friendly
No
47.308%
Yes
52.692%
Certification by country
The lowest 5 countries in AVG carbon footprint
Vietnam
China
Usa
Spain
India
Carbon footprint for every target audience
Kids
14.51%
Adults
19.18%
Household
32.41%
Business
33.90%

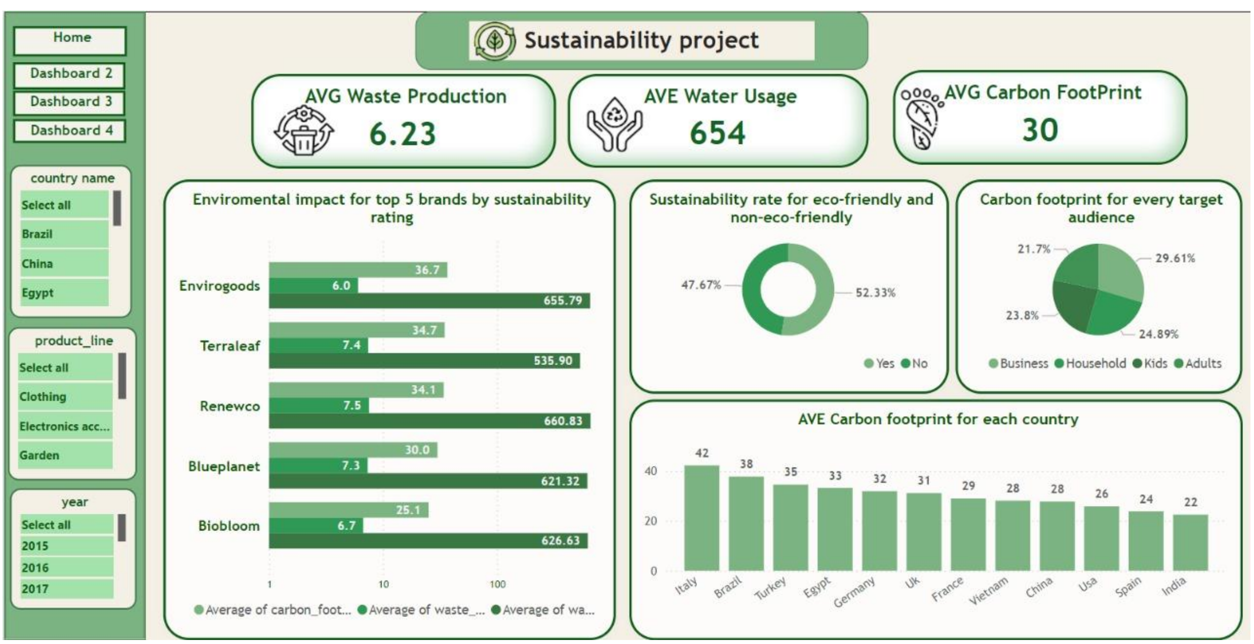

Sustainability Dashboard
environmental impact
Average water usage
Home
Das 1
Dash 3
Dash 4
Dash 5
Environmental impact of the top 5 countries in sustainability rating
Country Na..
Brazil
Egypt
Spain
Turkey
Uk
Avg. Carbon Footprint
Avg. Waste Production
Avg. Water Usage
Avg price for every brand category
21.74%
Startup
16.47%
Multinational
61.78%
Local
Environmental impacts over the years
Avg. Carbon Footprint
Avg. Waste Production
Avg. Water Usage
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Product line by sustainability rating
3.66
3.63
3.57
3.53
3.49
3.47
3.37
3.35
3.24
Personal care
Kitchen..
Home
Toys
Packaging
Office supplies
Electronic s accessor ies
Clothing
Garden
Distribution of material renewability
Partially Renewable
193
Fully Renewab
210
Not Renewable
176

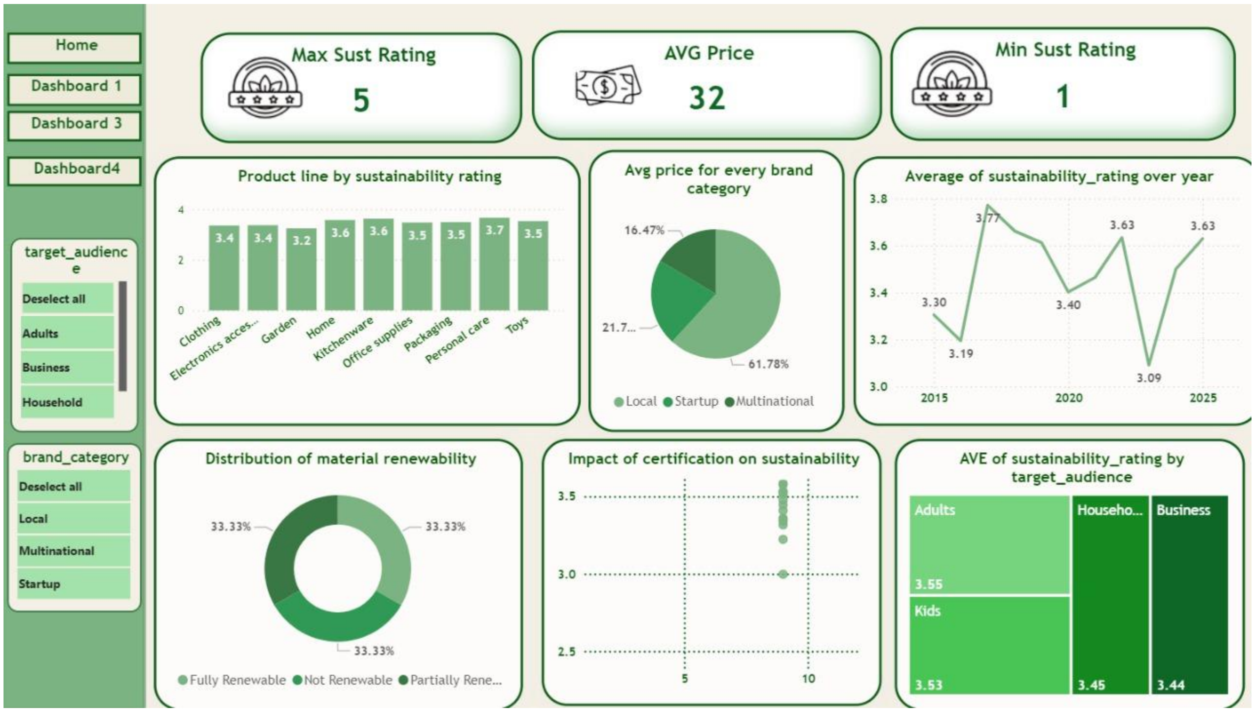

Sustainability Dashboard
enviromental impact
Average water usage
Home
Dash 1
Dash 2
Dash 4
Dash 5
Environmental impact of each product category
Product Line
Clothing 3.3542
Electronics accessories 3.3669
Garden 3.2439
Home 3.5711
Kitchenware 3.6258
Office supplies 3.4741
Packaging 3.4907
Personal care 3.6623
Toys 3.5305
Avg. Sustainability Rating
The bottom 5 brands in AVG sustainability rate
Renewco 3.35
Purecycle 3.34
Terraleaf 3.32
Ecosense 3.22
Blueplanet 3.00
Impact of certification on sustainability
Sustainability Rating
Count of Certification
AVG Sustainability by target audience
Adults
Household
Business
Kids
Distribution of AVG Sustainability Rating by Brand Category
32.448% Stable
30.832% Declining
36.720% Rising

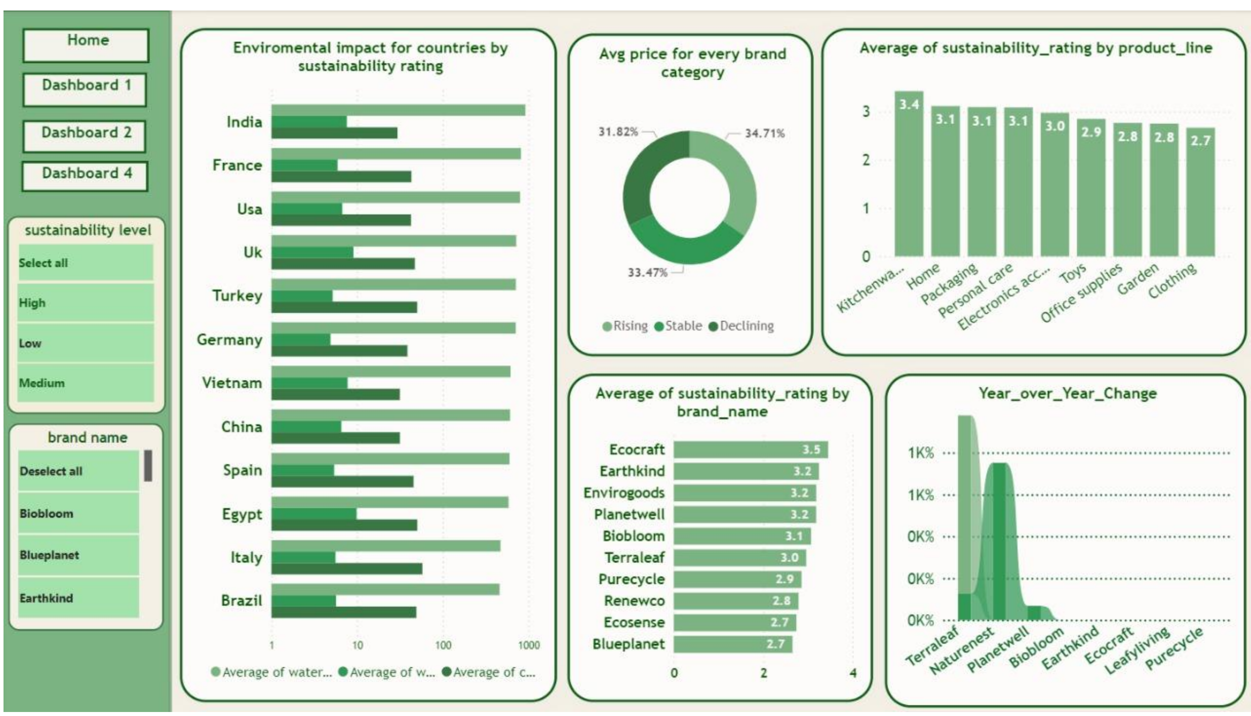

Sustainability Dashboard
Home
Dash 1
Dash 2
Dash 3
Dash 5
Average price by product line
Product Line
Clothing 34.034
Electronics accessories 33.273
Garden 27.797
Home 35.018
Kitchenware 30.975
Office supplies 31.646
Packaging 31.905
Personal care 34.889
Toys 29.538
Avg. Average Price
Brands with Recycling programs
Renewco
Blueplanet
Terraleaf
Planetwell
Envirogoods
Biobloom
Earthkind
Naturenest
Ecosense
Leafyliving
Earthly
Greenworks
Greenwave
Ecocraft
Sustainify
Ecomate
Purecycle
Greenroots
Verdevibe
Ecopure
Sustainability for every Material Type
Stainless steel
Glass
Bioplastic
Aluminum
Recycled plastic
Hemp
Organic cotton
Bamboo
Top 5 sustainable brands
Avg. Sustainability Rating
Greenroots 3.88
Planetwell 3.85
Ecopure 3.82
Ecomate 3.75
Ecocraft 3.74
Correlation between AVG price and AVG Sustainability
Avg. Average Price
Sustainability Rating

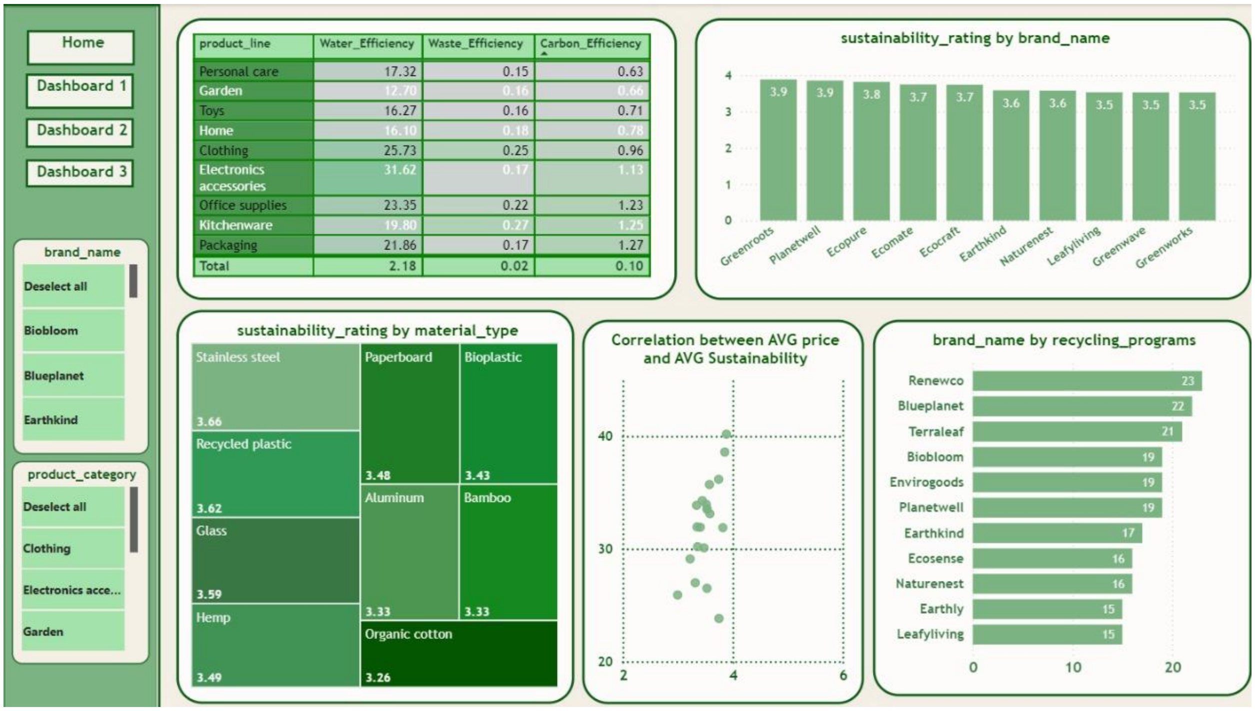

Sustainability Dashboard
Home
Dash 1
Dash 2
Dash 3
Dash 4
Year Over Year Change for brands
Brand Name
Blueplanet 38% 15%
Earthkind -13% -14%
Earthly -33% 15%
Ecocraft 31% -14%
Ecomate -31% -60%
Ecopure -60% -35%
Ecosense 263% 165%
Envirogoods 35% 57%
Greenroots -83% -53%
Greenwave 257% 94%
Greenworks -37% -20%
Leafyliving 35% -4%
Naturenest 34% 59%
Planetwell 52% -31%
Purecycle -51% -33%
Renewco 124% 134%
Sustainify -61% -39%
Terraleaf 138% 21%
Verdevibe -69% -57%
Year-over-Year Change in Carbon Footprint
Year-over-Year Change in Water Usage
Waste ,Water ,Carbon efficiency per product line
Product Line | Carbon Efficiency | Waste Efficiency | Water Efficiency
Clothing 96% 25% 12%
Electronics accessories 113% 17% 14%
Garden 66% 16% 6%
Home 78% 18% 7%
Kitchenware 125% 27% 9%
Office supplies 123% 22% 11%
Packaging 127% 17% 10%
Personal care 63% 15% 8%
Toys 71% 16% 7%

6.3.3 Power BI:


Sustainability project
Home
Dashboard 2
Dashboard 3
Dashboard 4
country name
Select all
Brazil
China
Egypt
product_line
Select all
Clothing
Electronics acc...
Garden
year
Select all
2015
2016
2017
AVG Waste Production
6.23
AVE Water Usage
654
AVG Carbon FootPrint
30
Enviromental impact for top 5 brands by sustainability rating
Envirogoods 36.7 6.0 655.79
Terraleaf 34.7 7.4 535.90
Renewco 34.1 7.5 660.83
Blueplanet 30.0 7.3 621.32
Biobloom 25.1 6.7 626.63
Average of carbon_foot... Average of waste_... Average of wa...
Sustainability rate for eco-friendly and non-eco-friendly
47.67%
52.33%
Yes No
Carbon footprint for every target audience
21.7%
29.61%
23.8%
24.89%
Business Household Kids Adults
AVE Carbon footprint for each country
Italy 42
Brazil 38
Turkey 35
Egypt 33
Germany 32
Uk 31
France 29
Vietnam 28
China 28
Usa 26
Spain 24
India 22


Home
Dashboard 1
Dashboard 3
Dashboard4
Max Sust Rating
5
AVG Price
32
Min Sust Rating
1
target_audience
Deselect all
Adults
Business
Household
brand_category
Deselect all
Local
Multinational
Startup
Product line by sustainability rating
3.4 3.4 3.2 3.6 3.6 3.5 3.5 3.7 3.5
Clothing
Electronics acces...
Garden
Home
Kitchenware
Office supplies
Packaging
Personal care
Toys
Avg price for every brand category
16.47%
21.7...
61.78%
Local Startup Multinational
Average of sustainability_rating over year
3.77 3.63 3.63 3.30 3.40 3.19 3.09
2015 2020 2025
Distribution of material renewability
33.33% 33.33% 33.33%
Fully Renewable Not Renewable Partially Rene...
Impact of certification on sustainability
3.5 3.0 2.5
5 10
AVE of sustainability_rating by target_audience
Adults 3.55
Kids 3.53
Househo... 3.45
Business 3.44


Home
Dashboard 1
Dashboard 2
Dashboard 4
sustainability level
Select all
High
Low
Medium
brand name
Deselect all
Biobloom
Blueplanet
Earthkind
Enviromental impact for countries by sustainability rating
India
France
Usa
Uk
Turkey
Germany
Vietnam
China
Spain
Egypt
Italy
Brazil
1 10 100 1000
Average of water... Average of w... Average of c...
Avg price for every brand category
31.82% 34.71% 33.47%
Rising Stable Declining
Average of sustainability_rating by product_line
3.4 3.1 3.1 3.1 3.0 2.9 2.8 2.8 2.7
Kitchenwa...
Home
Packaging
Personal care
Electronics acc...
Toys
Office supplies
Garden
Clothing
Average of sustainability_rating by brand_name
Ecocraft 3.5
Earthkind 3.2
Envirogoods 3.2
Planetwell 3.2
Biobloom 3.1
Terraleaf 3.0
Purecycle 2.9
Renewco 2.8
Ecosense 2.7
Blueplanet 2.7
Year_over_Year_Change
Terraleaf
Naturenest
Planetwell
Biobloom
Earthkind
Ecocraft
Leafyliving
Purecycle



7. Recommendations

Based on the quantitative insights derived from the **SQL**-structured datasets and visualized through **PowerBI** and **Tableau** dashboards, the following strategic recommendations are proposed:

7.1 Strategic Material Transition

- **Observation:** The MaterialDim data highlights renewable resources like Bamboo, Hemp, and Organic Cotton versus non-renewables like Stainless Steel.
- **Recommendation:** Manufacturers producing "Partially Sustainable" goods (Total Carbon: 62,505) should aggressively transition raw material sourcing toward materials flagged as Renewable: Yes. Prioritizing **Bioplastics** and **Organic Cotton** over conventional options can bridge the gap toward "Fully Sustainable" status.

7.2 Regional Optimization Protocols

- **Observation:** The geospatial analysis performed in **Tableau** highlights that the Americas and Asia struggle with high water usage averages (>850L).
- **Recommendation:** Implement **Water Stewardship Initiatives** specifically for supply chains in the Americas and Asia. Technologies such as closed-loop water systems should be piloted in these regions first to bring their averages in line with Europe's efficiency standards.

7.3 Certification as a Standard

- **Observation:** Brands such as *Biobloom*, *Blueplanet*, and *Earthkind* show varied distributions of sustainable vs. non-sustainable products.
- **Recommendation:** Standardize the adoption of high-impact certifications found in the CertificationDim (e.g., **GOTS**, **Fair Trade**, **ISO14001**) across all product lines. Brands should aim to eliminate "Not Sustainable" SKUs entirely, as they disproportionately skew the total environmental footprint.

7.4 Operationalize BI Dashboards for Real-Time Monitoring

- **Observation:** The current insights are based on a snapshot analysis.
- **Recommendation:** Transition the **PowerBI** and **Tableau** visualizations created for this project into live, automated dashboards linked directly to the SQL data warehouse. This will enable real-time monitoring of Carbon Footprint and Water Usage KPIs, allowing stakeholders to detect sustainability rating dips immediately rather than retrospectively.

8. Conclusion

The comprehensive analysis of the sustainability dataset yields several critical conclusions regarding the current state of eco-friendly manufacturing:

8.1 The "Green Gap" in Carbon Emissions

The most significant finding is the drastic correlation between sustainability classification and carbon output.

- **Not Sustainable** products accounted for a total carbon footprint of approximately **94,300 units**.
- **Fully Sustainable** products accounted for only **17,929 units**.
- **Conclusion:** Fully sustainable products generate approximately **81% less total carbon emissions** than their non-sustainable counterparts within this dataset. Transitioning production lines to "Fully Sustainable" standards offers the highest leverage for decarbonization.

8.2 Regional Efficiency Disparities

Data regarding average resource intensity reveals distinct regional trends:

- **Europe** demonstrates the highest efficiency, with the lowest average carbon footprint per unit (**33.88**).
- **Africa** and the **Americas** show higher average carbon footprints (**36.98** and **36.45**, respectively).
- **Water Usage:** The Americas and Asia recorded the highest average water usage (approx. 860L and 857L), suggesting that water stewardship is a critical challenge in these specific regions compared to Europe (817L).

8.3 Market Trend Correlations

The analysis of recycling programs indicates a strong link between market momentum and environmental responsibility. Brands operating in "Rising" market trends show a higher adoption or count of recycling programs (1,696 instances) compared to those in "Declining" trends (1,665 instances). This suggests that consumer demand is currently driving corporate adoption of circular economy principles.

9. Team Performance:

9..1. Excel:

- Doaa: Extracting insights and analyzing KPIs.
- Esraa: Creating Pivot Tables and designing charts.
- Dina: Data cleaning, modeling, and generating insights.
- Nour: Conducting statistical analysis.
- George: Designing a full interactive dashboard.
- Malak: Creating a complete dashboard.

9..2. SQL:

- Doaa: Inserting data
- Esraa: Creating the database and empty tables, inserting raw data.
- Dina: performing cleaning, Data modeling, querying, & extracting insights.

9..3. Python:

- Doaa: Extracting insights.
- Esraa: Extracting insights.
- George: Extracting insights.
- Dina: Extracting insights, & Data cleaning and modeling.
- Malak: Creating a dashboard.

9..4. Tableau:

- Dina: Data modeling and insights.
- Doaa: Insights extraction.
- Malak: Designing a dashboard.

9..5. Power BI:

- Dina: Data modeling and insights.
- Doaa: Designing a dashboard.

10.References:

- From Kaggle
- https://sustainability-dashboard-c9huhdudy3xhfhktvvvzsw.streamlit.app/
- https://github.com/DEPIProjects/Sustainability

11.Appendices:

11..1. Some Python Analysis:

```
# =============== Insight 4: Sustainability rating trend over the years =
trend = df.groupby("year")["sustainability_rating"].mean()

print("\n Sustainability Rating Trend Over the Years:")
print(trend)

plt.figure()
sns.lineplot(x=trend.index, y=trend.values, marker="o", color="green")
plt.title("Sustainability Rating Over Time")
plt.xlabel("Year")
plt.ylabel("Average Sustainability Rating")
plt.tight_layout()
plt.show()
```

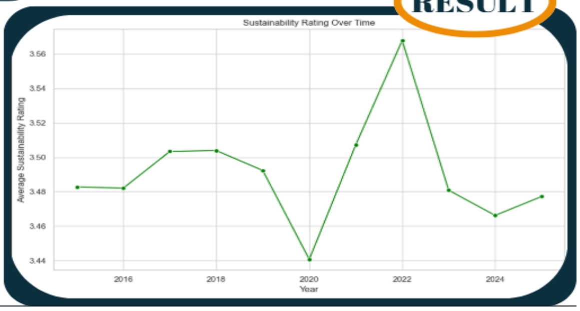
CODE

Python

```
 Sustainability Rating Trend Over the Years:
year
2015    3.482790
2016    3.482092
2017    3.503412
2018    3.503910
2019    3.492336
2020    3.440736
2021    3.507171
2022    3.567837
2023    3.480845
2024    3.466258
2025    3.477226
Name: sustainability_rating, dtype: float64
```

RESULT

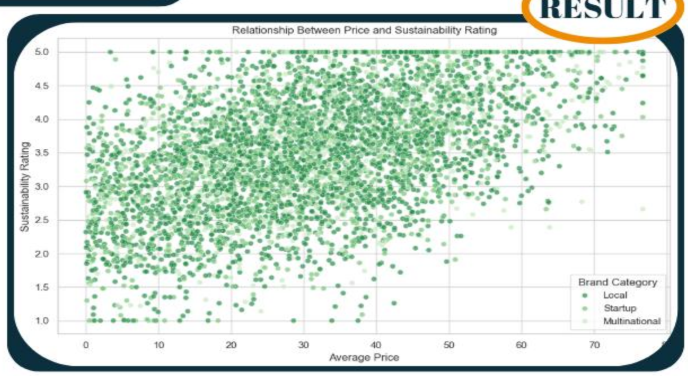
RESULT




```
# =============== Insight 3: Relationship between price and sustainability rating ==
plt.figure()
sns.scatterplot(data=df, x="average_price", y="sustainability_rating",
                hue="brand_category", alpha=0.7,palette="Greens_r")
plt.title("Relationship Between Price and Sustainability Rating")
plt.xlabel("Average Price")
plt.ylabel("Sustainability Rating")
plt.legend(title="Brand Category")
plt.tight_layout()
plt.show()
```

CODE

RESULT



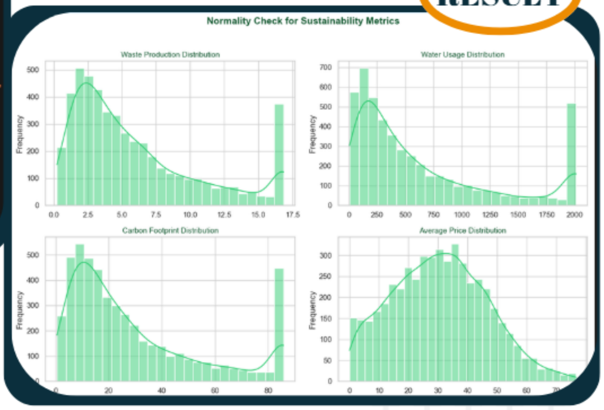
Python

```python
# Normality Check for Sustainability Metrics
main_green = '#27ae60'
light_green = '#2ecc71'
dark_green = '#145a32'

cols_to_check = ['waste_production', 'water_usage',
                 'carbon_footprint', 'average_price']
# Create 2x2 subplots for normality visualizations
fig, axes = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle("Normality Check for Sustainability Metrics",
             fontsize=14, color=dark_green, fontweight='bold')

for i, col in enumerate(cols_to_check):
    row, col_idx = divmod(i, 2)

    # Histogram with KDE (density curve)
    sns.histplot(df[col], kde=True, color=light_green, ax=axes[row, col_idx])
    axes[row, col_idx].set_title(f"{col.replace('_', ' ').title()} Distribution"
                                 , color=dark_green, fontsize=11)
    axes[row, col_idx].set_xlabel("")
    axes[row, col_idx].set_ylabel("Frequency")

plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()
```

CODE

Python

RESULT

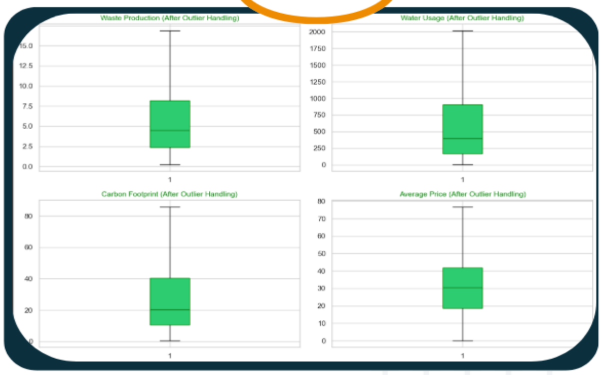



```python
# Create a figure with 4 subplots (2 rows x 2 columns)
fig, axes = plt.subplots(2, 2, figsize=(12, 8))

# Define a green color palette
box_color = '#2ecc71'

# Plot each variable after handling outliers
axes[0, 0].boxplot(df['waste_production'], patch_artist=True,
                   boxprops=dict(facecolor=box_color,
                                 color='green'),
                   medianprops=dict(color='darkgreen'))
axes[0, 0].set_title('Waste Production (After Outlier Handling)',
                     fontsize=11, color='green')

axes[0, 1].boxplot(df['water_usage'], patch_artist=True,
                   boxprops=dict(facecolor=box_color, color='green'),
                   medianprops=dict(color='darkgreen'))
axes[0, 1].set_title('Water Usage (After Outlier Handling)',
                     fontsize=11, color='green')

axes[1, 0].boxplot(df['carbon_footprint'], patch_artist=True,
                   boxprops=dict(facecolor=box_color, color='green'),
                   medianprops=dict(color='darkgreen'))
axes[1, 0].set_title('Carbon Footprint (After Outlier Handling)',
                    fontsize=11, color='green')

axes[1, 1].boxplot(df['average_price'], patch_artist=True,
                   boxprops=dict(facecolor=box_color, color='green'),
                   medianprops=dict(color='darkgreen'))
axes[1, 1].set_title('Average Price (After Outlier Handling)',
                    fontsize=11, color='green')

plt.tight_layout()
plt.show()
```

CODE

Python

RESULT



11..2. Some SQL Analysis:

11..3.

```sql
-- ========================================================
-- STEP 1: Compute total counts for quartile positions
-- ========================================================
SET @total_sus = (SELECT COUNT(*) FROM raw_data WHERE sustainability_rating IS NOT NULL);
SET @q1_sus_pos = FLOOR(@total_sus * 0.25);
SET @q3_sus_pos = FLOOR(@total_sus * 0.75);

SET @total_price = (SELECT COUNT(*) FROM raw_data WHERE average_price IS NOT NULL);
SET @q1_price_pos = FLOOR(@total_price * 0.25);
SET @q3_price_pos = FLOOR(@total_price * 0.75);

SET @total_carbon = (SELECT COUNT(*) FROM raw_data WHERE carbon_footprint IS NOT NULL);
SET @q1_carbon_pos = FLOOR(@total_carbon * 0.25);
SET @q3_carbon_pos = FLOOR(@total_carbon * 0.75);

SET @total_water = (SELECT COUNT(*) FROM raw_data WHERE water_usage IS NOT NULL);
SET @q1_water_pos = FLOOR(@total_water * 0.25);
SET @q3_water_pos = FLOOR(@total_water * 0.75);

SET @total_waste = (SELECT COUNT(*) FROM raw_data WHERE waste_production IS NOT NULL);
SET @q1_waste_pos = FLOOR(@total_waste * 0.25);
SET @q3_waste_pos = FLOOR(@total_waste * 0.75);
```

```sql
-- STEP 8: Verification summary  Before Vs After
-- ========================================================
SELECT
  SUM(sus_outlier) AS sus_outliers_before,
  SUM(price_outlier) AS price_outliers_before,
  SUM(carbon_outlier) AS carbon_outliers_before,
  SUM(water_outlier) AS water_outliers_before,
  SUM(waste_outlier) AS waste_outliers_before
FROM outliers_before;

-- Checking for handling outliers
SELECT
  SUM(sus_outlier) AS sus_outliers_after,
  SUM(price_outlier) AS price_outliers_after,
  SUM(carbon_outlier) AS carbon_outliers_after,
  SUM(water_outlier) AS water_outliers_after,
  SUM(waste_outlier) AS waste_outliers_after
FROM outliers_after;
```

SQL

```sql
-- STEP 4: Detect outliers before Winsorizing
-- ========================================================
DROP TEMPORARY TABLE IF EXISTS outliers_before;
CREATE TEMPORARY TABLE outliers_before AS
SELECT *,
  CASE WHEN sustainability_rating IS NOT NULL AND (ROUND(sustainability_rating,4) < @sus_lower OR ROUND(sustainability_rating,4) > @sus_upper) THEN 1 ELSE 0 END AS sus_outlier,
  CASE WHEN average_price IS NOT NULL AND (ROUND(average_price,4) < @price_lower OR ROUND(average_price,4) > @price_upper) THEN 1 ELSE 0 END AS price_outlier,
  CASE WHEN carbon_footprint IS NOT NULL AND (ROUND(carbon_footprint,4) < @carbon_lower OR ROUND(carbon_footprint,4) > @carbon_upper) THEN 1 ELSE 0 END AS carbon_outlier,
  CASE WHEN water_usage IS NOT NULL AND (ROUND(water_usage,4) < @water_lower OR ROUND(water_usage,4) > @water_upper) THEN 1 ELSE 0 END AS water_outlier,
  CASE WHEN waste_production IS NOT NULL AND (ROUND(waste_production,4) < @waste_lower OR ROUND(waste_production,4) > @waste_upper) THEN 1 ELSE 0 END AS waste_outlier
FROM raw_data;
```

```sql
STEP 2: Compute Q1 and Q3 using PREPARE statements
-- ========================================================
-- sustainability_rating
SET @sql = CONCAT(
  'SELECT sustainability_rating INTO @Q1_sus FROM raw_data ',
  'WHERE sustainability_rating IS NOT NULL ORDER BY sustainability_rating LIMIT ', @q1_sus_pos, ',1
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

SET @sql = CONCAT(
  'SELECT sustainability_rating INTO @Q3_sus FROM raw_data ',
  'WHERE sustainability_rating IS NOT NULL ORDER BY sustainability_rating LIMIT ', @q3_sus_pos, ',1
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

-- average_price
SET @sql = CONCAT(
  'SELECT average_price INTO @Q1_price FROM raw_data ',
  'WHERE average_price IS NOT NULL ORDER BY average_price LIMIT ', @q1_price_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

SET @sql = CONCAT(
  'SELECT average_price INTO @Q3_price FROM raw_data ',
  'WHERE average_price IS NOT NULL ORDER BY average_price LIMIT ', @q3_price_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

-- carbon_footprint
SET @sql = CONCAT(
  'SELECT carbon_footprint INTO @Q1_carbon FROM raw_data ',
  'WHERE carbon_footprint IS NOT NULL ORDER BY carbon_footprint LIMIT ', @q1_carbon_pos, ',1'
);
EPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;
```

SQL

```sql
@sql = CONCAT(
  'SELECT carbon_footprint INTO @Q1_carbon FROM raw_data ',
  'WHERE carbon_footprint IS NOT NULL ORDER BY carbon_footprint LIMIT ', @q1_carbon_pos, ',1
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

SET @sql = CONCAT(
  'SELECT carbon_footprint INTO @Q3_carbon FROM raw_data ',
  'WHERE carbon_footprint IS NOT NULL ORDER BY carbon_footprint LIMIT ', @q3_carbon_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

-- water_usage
SET @sql = CONCAT(
  'SELECT water_usage INTO @Q1_water FROM raw_data ',
  'WHERE water_usage IS NOT NULL ORDER BY water_usage LIMIT ', @q1_water_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

SET @sql = CONCAT(
  'SELECT water_usage INTO @Q3_water FROM raw_data ',
  'WHERE water_usage IS NOT NULL ORDER BY water_usage LIMIT ', @q3_water_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

-- waste_production
SET @sql = CONCAT(
  'SELECT waste_production INTO @Q1_waste FROM raw_data ',
  'WHERE waste_production IS NOT NULL ORDER BY waste_production LIMIT ', @q1_waste_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;

SET @sql = CONCAT(
  'SELECT waste_production INTO @Q3_waste FROM raw_data ',
  'WHERE waste_production IS NOT NULL ORDER BY waste_production LIMIT ', @q3_waste_pos, ',1'
);
PREPARE stmt FROM @sql; EXECUTE stmt; DEALLOCATE PREPARE stmt;
```

```sql
-- Product categories where AVG carbon_footprint < 34
SELECT
    product_category,
    ROUND(AVG(carbon_footprint),2) AS average_carbon_footprint
FROM product_category_dim AS pdm
LEFT JOIN sustainability_fact AS sf
    ON pdm.product_category_id = sf.product_category_id
GROUP BY product_category
HAVING ROUND(AVG(carbon_footprint),2) < 34  -- average = 34.59
ORDER BY average_carbon_footprint;
```

```sql
-- Top 3 regions by average sustainability
SELECT DISTINCT
    region,
    ROUND(AVG(sustainability_rating), 3) AS average_sustianability
FROM sustainability_fact AS sf
INNER JOIN country_dim AS c
    ON sf.country_id = c.country_id
GROUP BY region
ORDER BY average_sustianability DESC
LIMIT 3;
```

SQL

```sql
-- Top 5 countries by average sustainability rating
SELECT DISTINCT
    c.country_name,
    ROUND(AVG(sustainability_rating), 3) AS average_sustainability
FROM sustainability_fact AS sf
INNER JOIN country_dim AS c
    ON sf.country_id = c.country_id
GROUP BY country_name
ORDER BY average_sustainability DESC
LIMIT 5;
```

```sql
-- Bottom 5 brands (lowest avg rating)
SELECT
    b.brand_name,
    ROUND(AVG(sf.sustainability_rating), 3) AS avg_rating
FROM sustainability_fact AS sf
JOIN brand_dim b
    ON sf.brand_id = b.brand_id
GROUP BY b.brand_name
ORDER BY avg_rating ASC
LIMIT 5;
```

```sql
-- Waste production by material type (avg)
SELECT
    m.material_type,
    ROUND(AVG(f.waste_production),2) AS avg_waste
FROM sustainability_fact f
JOIN material_dim m
    ON f.material_id = m.material_id
GROUP BY m.material_type
ORDER BY avg_waste ASC;
```

SQL

```sql
-- Number of certifications per brand category
SELECT
    brand_category,
    COUNT(DISTINCT certification) AS number_of_certifications
FROM brand_dim AS b
INNER JOIN sustainability_fact AS sf
    ON b.brand_id = sf.brand_id
INNER JOIN certification_dim AS c
    ON sf.certification_id = c.certification_id
GROUP BY brand_category;
```

```sql
-- ==============================================
-- Material types that are fully renewable
SELECT material_type
FROM material_dim
WHERE material_status = 'Fully Renewable';
-- ==============================================
-- Material types that are partially renewable
SELECT material_type
FROM material_dim
WHERE material_status = 'Partially Renewable';
-- ==============================================
-- Material types that are not renewable
SELECT material_type
FROM material_dim
WHERE material_status = 'Not Renewable';
-- ==============================================
```